

Andrew McDonagh

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EDUCATION

University of Wisconsin–Madison | Madison, WI

May 2026

Bachelor of Science in Computer Science

IFSA Tech Career Accelerator | Prague, Czech Republic

January 2025 – May 2025

Semester Abroad – Cybersecurity & Software Engineering Focus

Loyola Academy | Wilmette, IL

August 2018 – May 2022

High School Diploma | Summa Cum Laude, 4-Year First Honor Roll, 4-Year Varsity Lacrosse

RELEVANT COURSEWORK

CS 542: Software Security | UW–Madison

- Studied and applied mitigations for vulnerabilities spanning nearly all of the OWASP Top 25, including SQL/command/XML/code injection, XSS, CSRF, directory traversal, serialization flaws, and session management.
- Performed hands-on web application penetration testing using OWASP ZAP (Zed Attack Proxy), including request tampering, injection attacks, and CORS exploitation.
- Participated in a tabletop cybersecurity exercise alongside an FBI agent, Alliant Energy personnel, and industry SMEs, focusing on OT/ICS threat response and mitigation strategies.

Computer and Network Security | IFSA

- Configured firewalls, VPNs, and secure network protocols; analyzed vulnerabilities across local and cloud infrastructure and applied encryption and access control principles through hands-on labs.
- Implemented OS fundamentals in Linux: process scheduling, virtual memory, file I/O, IPC, and concurrency control.

CS 540: Intro to Artificial Intelligence | UW–Madison

- Applied foundational AI/ML techniques including neural network training, NLP, game theory, and search algorithms.

PROJECTS

RAG System | Hexagon | IFSA Capstone

January 2025 – May 2025

- Built a Retrieval-Augmented Generation (RAG) system to capture and surface tacit engineering knowledge at risk of being lost as senior Hexagon engineers retire.
- Served as primary backend developer for the LLM pipeline; designed retrieval logic and managed data ingestion.
- Developed with LangChain; implemented Text-to-SQL query capability enabling natural language interaction with structured engineering data.

TimeCoin | CS 506 | UW–Madison

January 2026 – May 2026

- Collaborating on a team of developers to design and build a cryptocurrency platform from the ground up, covering blockchain logic, transaction handling, and wallet management.
- Focused on backend architecture and database design using Java and MySQL; integrated Docker containerization and GitLab CI/CD for automated build and deployment pipelines.
- Applied REST API design patterns and followed TDD practices with JUnit and Mockito throughout development.

EXPERIENCE

Beta Theta Pi – UW–Madison | Madison, WI

May 2023 – May 2024

House Manager

- Oversaw all facility operations for a large chapter house, managing vendors and communicating with alumni in the Beta Building Association to resolve maintenance priorities.

SKILLS & CERTIFICATIONS

Languages: Java, Python, JavaScript, Bash, C, R, SQL

Frameworks & Tools: LangChain, JUnit, Mockito, OWASP ZAP, Microsoft Threat Modeling Tool (MTML)

DevOps & Infrastructure: Docker, GitLab CI/CD, GitHub, Linux, Git, MySQL, REST APIs

Operating Systems: Linux (Ubuntu/Kali), macOS, Windows

Certifications: Google Cybersecurity Professional Certificate (July 2025) | CompTIA Security+ (In Progress)

Interests: Incident Response, Digital Forensics, Threat Intelligence, Software Engineering